WORKED SOLUTION

STEP II 1993, Question 6

Pure • Matrix Algebra Counterexamples

STEP II 1993, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q6

This question has a full guided treatment in *Beyond STEP*, Section 7.1, *Rules That Fail for Matrices*: https://beyondstep.keithmcnulty.org/ch07/matrix-rules.html

The key idea

Each assertion is something that would be true for real numbers, so the question is really asking which habits from ordinary algebra survive for matrices. Two things break: matrix multiplication need not commute, and a product of two non-zero matrices can be zero. A single well-chosen pair of very simple matrices settles (i) and (ii), the adjugate matrix settles (iii), and a repeated "root" with a nilpotent twist settles (iv).

Solution

Throughout, $\mathbf{I}$ is the 2×2 identity and $\mathbf{0}$ the zero matrix.

Part (i)

False. Take

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

both non-zero. Then

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \mathbf{BA} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \neq \mathbf{0}.$$

Ask yourself How would you find a counterexample like this, rather than just guess one?

We want $\mathbf{AB} = \mathbf{0}$, so every column of $\mathbf{B}$ must be killed by $\mathbf{A}$. The matrix $\mathbf{A}$ above kills any vector of the form $(0, y)^{\mathsf{T}}$, so we may take the columns of $\mathbf{B}$ to be of that form.

Then we just check whether $\mathbf{BA}$ happens to vanish; with the most obvious choice it doesn't.

Part (ii)

False. Expanding carefully, keeping the order of every product,

$$(\mathbf{A}-\mathbf{B})(\mathbf{A}+\mathbf{B}) = \mathbf{A}^2 + \mathbf{AB} - \mathbf{BA} - \mathbf{B}^2,$$

so the assertion holds exactly when $\mathbf{AB} = \mathbf{BA}$. The matrices of part (i) don't commute, so they give a counterexample: there $\mathbf{AB} - \mathbf{BA} = -\mathbf{BA} \neq \mathbf{0}$. Explicitly,

$$(\mathbf{A}-\mathbf{B})(\mathbf{A}+\mathbf{B}) = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \qquad \mathbf{A}^2 - \mathbf{B}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

using $\mathbf{A}^2 = \mathbf{A}$ and $\mathbf{B}^2 = \mathbf{0}$.

Part (iii)

True. We prove both directions.

If $\mathbf{AX} = \mathbf{0}$ *has a non-zero solution, then* $\det \mathbf{A} = 0$. We prove the contrapositive. Suppose $\det \mathbf{A} \neq 0$. Then $\mathbf{A}^{-1}$ exists, and $\mathbf{AX} = \mathbf{0}$ gives $\mathbf{X} = \mathbf{A}^{-1}\mathbf{AX} = \mathbf{A}^{-1}\mathbf{0} = \mathbf{0}$. So the only solution is the zero matrix.

If $\det \mathbf{A} = 0$, *then there is a non-zero solution.* Write

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

(so $\mathbf{X}$ is the adjugate of $\mathbf{A}$). Multiplying out,

$$\mathbf{AX} = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = (\det \mathbf{A})\,\mathbf{I} = \mathbf{0}.$$

The entries of $\mathbf{X}$ are those of $\mathbf{A}$ rearranged, with some signs changed, and $\mathbf{A} \neq \mathbf{0}$, so $\mathbf{X} \neq \mathbf{0}$. This completes the proof. Notice that the hypothesis $\mathbf{A} \neq \mathbf{0}$ is used here: it is what guarantees that the adjugate is non-zero.

Part (iv)

False. For numbers, a quadratic has at most two roots, but the proof of that fact uses "if $pq = 0$ then $p = 0$ or $q = 0$", which part (i) has just shown fails for matrices. So look for a quadratic that factorises as a perfect square. Take $\mathbf{A} = -2\mathbf{I}$ and $\mathbf{B} = \mathbf{I}$, both non-zero. Since $\mathbf{I}$ commutes with everything,

$$\mathbf{X}^2 - 2\mathbf{X} + \mathbf{I} = (\mathbf{X} - \mathbf{I})^2,$$

so we need $(\mathbf{X} - \mathbf{I})^2 = \mathbf{0}$. Any matrix $\mathbf{N}$ with $\mathbf{N}^2 = \mathbf{0}$ will do for $\mathbf{X} - \mathbf{I}$, and there are infinitely many. For each real t let

$$\mathbf{X}_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

Then

$$\mathbf{X}_t^2 - 2\mathbf{X}_t + \mathbf{I} = \begin{pmatrix} 1 & 2t \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 2 & 2t \\ 0 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{0}.$$

Each $\mathbf{X}_t$ is non-zero, and different values of t give different matrices. So this equation has infinitely many solutions, not at most two.

? **Ask yourself** Is there anything special about $\mathbf{A} = -2\mathbf{I}$, $\mathbf{B} = \mathbf{I}$?

Not really. With $\mathbf{A} = -2\lambda\mathbf{I}$ and $\mathbf{B} = \lambda^2\mathbf{I}$ (for any $\lambda \neq 0$), every $\mathbf{X} = \lambda\mathbf{I} + \mathbf{N}$ with $\mathbf{N}^2 = \mathbf{0}$ works. The nilpotent matrices themselves form a large family: $\begin{pmatrix} pq & -p^2 \\ q^2 & -pq \end{pmatrix}$ squares to $\mathbf{0}$ for all real p, q. The essential point is only that the quadratic factorises as a product of two matrices, and such a product can vanish in many more ways than for numbers.

Checks

Multiply out each counterexample by hand: in (i), $\mathbf{AB}$ has every entry 0 while $\mathbf{BA}$ has a 1 in the bottom-left corner. The identity $(\mathbf{A}-\mathbf{B})(\mathbf{A}+\mathbf{B}) = \mathbf{A}^2 - \mathbf{B}^2 + \mathbf{AB} - \mathbf{BA}$ was tested on 1000 random integer pairs, and $\mathbf{A}\,\mathrm{adj}\,\mathbf{A} = \mathbf{0}$ with $\mathrm{adj}\,\mathbf{A} \neq \mathbf{0}$ on 2000 random singular matrices. In (iv), try $t = 3$: $\mathbf{X}_3^2 = \begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix}$, and $\mathbf{X}_3^2 - 2\mathbf{X}_3 + \mathbf{I} = \mathbf{0}$.

Common slips

- Expanding $(\mathbf{A}-\mathbf{B})(\mathbf{A}+\mathbf{B})$ as if $\mathbf{AB} = \mathbf{BA}$. Writing "false" in (ii) without an explicit counterexample earns little: the question asks for one.
- In (iii), proving only one direction of the "if and only if".
- In (iv), "solving" the quadratic with the formula or by factorising and concluding $\mathbf{X} = \mathbf{I}$ is the only root. Factorising is fine, but $\mathbf{P}^2 = \mathbf{0}$ does not imply $\mathbf{P} = \mathbf{0}$.

Beyond STEP

The 2×2 real matrices form a *non-commutative ring with zero divisors*, and each part of this question tests a law that holds in a field but fails in such a ring. Part (iii) is the 2×2 case of a central theorem of linear algebra: a square matrix has a non-trivial null space exactly when its determinant vanishes. Matrices with $\mathbf{N}^2 = \mathbf{0}$, like those in (iv), are called nilpotent and are the key ingredient in the Jordan normal form. Strang's *Introduction to Linear Algebra* covers the linear algebra, and Allenby's *Rings, Fields and Groups* the ring theory.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.